

ASSIGNMENT 6 — Windows, Tabs and Iframes

1. Experiment Title

Handling Windows, Tabs and Iframes Using Selenium

2. Problem Statement

Automate a web page containing an embedded iframe and a mechanism for opening a new browser tab/window. The script should interact with the iframe, switch to the newly opened tab, retrieve its title, close it, and return to the original browser context.

3. Objective

- To switch into an iframe using Selenium.
- To interact with elements inside an iframe.
- To handle multiple browser windows/tabs.
- To use `window_handles` to identify browser contexts.
- To switch between the main window and new tab.
- To retrieve the title of the new tab and close it.

4. Tools, Software and Concepts Used

Tools & Software:

- Python
- Selenium WebDriver
- Google Chrome
- VS Code / PyCharm

Concepts:

- Iframes
- Browser Windows
- Browser Tabs
- `switch_to.frame()`

- switch_to.default_content()
- window_handles
- switch_to.window()
- Browser Title
- Selenium WebDriver

5. Source Code / Implementation Steps

```

from selenium import webdriver

from selenium.webdriver.common.by import By

import time

def test_windows_tabs_and_iframes():

    driver = webdriver.Chrome()

    driver.maximize_window()

    driver.get(

"https://www.selenium.dev/selenium/web/window_switching_tests/page_with_f
rame.html"

    )

    main_window = driver.current_window_handle

    iframe = driver.find_element(By.TAG_NAME, "iframe")

    driver.switch_to.frame(iframe)

    iframe_text = driver.find_element(By.TAG_NAME, "body").text

    print("Iframe content:", iframe_text)

    assert len(iframe_text) > 0

    driver.switch_to.default_content()

    driver.find_element(By.LINK_TEXT, "Open new window").click()

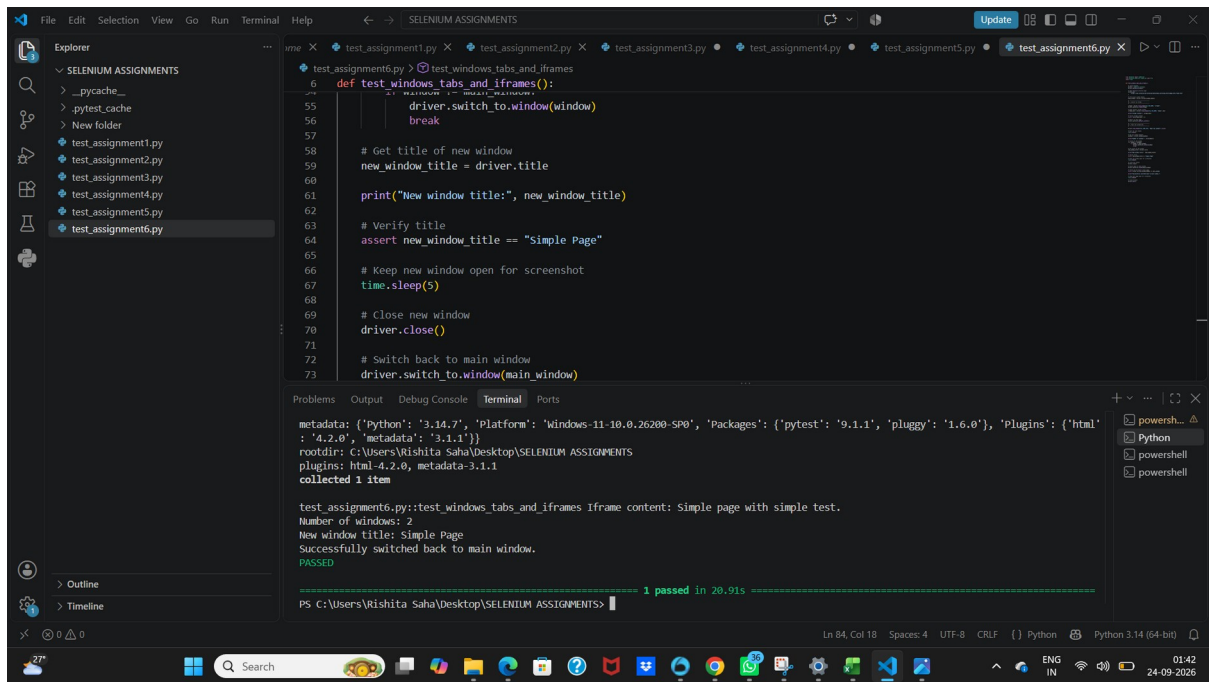
    time.sleep(2)

```

```
windows = driver.window_handles
print("Number of windows:", len(windows))
for window in windows:
    if window != main_window:
        driver.switch_to.window(window)
        break
new_window_title = driver.title
print("New window title:", new_window_title)
assert new_window_title == "Simple Page"
time.sleep(5)
driver.close()
driver.switch_to.window(main_window)
assert driver.current_window_handle == main_window

print("Successfully switched back to main window.")
time.sleep(5)
driver.quit()
```

6. Output with Clearly Labelled Screenshots



Successful Execution

Show the terminal after the complete script finishes.

7. Result

The iframe was successfully accessed and interacted with. A new browser tab was opened, its window handle was identified, the tab title was retrieved, the tab was closed, and control was returned to the main browser window.

8. Observation

- Selenium requires switching into an iframe before interacting with its elements.
- `switch_to.default_content()` returns control to the main document.
- `window_handles` provides the identifiers of open browser windows/tabs.
- `switch_to.window()` changes the current browser context.
- The new tab can be closed independently before returning to the original window.

9. Conclusion

The experiment successfully demonstrated handling of **iframes, browser windows, and tabs using Selenium WebDriver**. These techniques are useful for automating web applications containing embedded content and multiple browser contexts.